1. For each of the following examples, explain whether this is a case of external or internal economies of scale.
   a. Most major banks, financial firms (including hedge funds), and the stock market are located in New York City.
   b. Hartford, CT, is the insurance capital of the northeast.
   c. Close to half the world's wide body aircraft are assembled in Seattle.
   d. Most of the world's aluminum is smelted in Norway of Canada.

2. Using the data I provided in class, suppose that the two countries were to integrate their automobile market with a third country with an annual market of 3.6 million cars.
   a. Find the number of firms.
   b. The output per firm.
   c. The average cost and price per car in the new integrated market after trade.

3. The decline in the average cost of production as the *cumulative* output of the industry increases and firms accumulate knowledge (e.g., learning-by-doing) is called dynamic external economies. The equation for dynamic external economies—the so-called "learning curve"—can be expressed as $AC = \alpha Q^{\beta}$.
   a. Explain the economic meaning of the "learning curve."
   b. What is the economic meaning and sign of the parameters α and β? (Hint: take logs of the equation).

   Extra credit (1 pt.): What's the difference between an economist and a confused old man with Alzheimer's?

- To max π of the typical firm:

$$\pi = PX - C = P\,s\left[1/n - b(P-\bar{P})\right] - F - c\,s\left[1/n - b(P-\bar{P})\right] \quad (1)$$

$$\frac{d\pi}{dP} = 0 \Rightarrow \quad X - sbP + sbc = 0 \quad (2)$$

- Since all firms are symmetric, in equil. $P = \bar{P}$ and $X = s/n \Rightarrow \frac{s}{n} = sbP - sbc$

or $bP = \frac{1}{n} + bc$

or $$P = 1/bn + c \quad (3)$$

- Since $$AC = Fn/s + c \quad (4)$$

- In equil., the price charged by a typical firm must also equal its average cost, so

$$1/bn + c = Fn/s + c \quad (5)$$

$$\Rightarrow \quad n = \sqrt{s/bF} \quad (6)$$

- And the price charged by the representative firm is:

$$P = 1/bn + c = c + \sqrt{F/sb} \quad (7)$$

- Sales per firm:

$$X_i = s/n = \sqrt{sbF} \quad (8)$$

$$\Rightarrow X = s/\sqrt{\frac{s}{bF}} \quad \text{or} \quad X^2 = \frac{s^2}{\frac{s}{bF}}$$

$$\text{or} \quad X^2 = sbF \Rightarrow X = \sqrt{sbF}$$

Numerical example (2)

Homes:

- $P = \$5000 + 1/[(1/30{,}000) \times 6] = \$5000 + \$5000$

$P = \$10{,}000$; $X_i = \$150{,}000 = 900{,}000/6$; $\boxed{S = 900{,}000}$

$[P = c + 1/(b \times n)]$

- $AC = (\$750{,}000{,}000/150{,}000) + \$5000 = \$10{,}000$

$F = \$750{,}000{,}000$; $b = 1/30{,}000$; $c = \$5000$; $n = \sqrt{\frac{900{,}000}{25{,}000}}$

$S = \$900{,}000$

$n = \sqrt{36} = 6$

Foreign: 1.6 million = S

$$n = \sqrt{\frac{1{,}600{,}000}{25{,}000}} = \sqrt{64} = 8$$

$$\boxed{F \times b = \frac{750{,}000{,}000}{30{,}000} = 25{,}000}$$

$$\frac{1{,}600{,}000}{8} = 200{,}000 = X_i$$

$$\Rightarrow AC = (\$750{,}000{,}000/200{,}000) + \$5000 = \$3750 + \$5000$$

$$AC = \$8{,}750$$

$$P = 8000 = \$5000 + 1/[(1/30{,}000) \times 8] = \$8750$$

Suppose H and F trade autos in one large market (integrated)

$n = \sqrt{\frac{2{,}500{,}000}{25{,}000}} = 10$

$n = 10$; $S = 2.5$ million autos, w/ each firm producing 250,000.

$P = \$5000 + 1/[(1/30{,}000) \times 10] = \$5000 + \$3000 = \$8{,}000$

$AC = (\$750{,}000{,}000/250{,}000) + \$5000 = \$8{,}000$

(over →)